Layer 0: Wound Biology

Protease Cascade ODE: MMP-9 and HNE in Chronic Wound Fluid

Eton iGEM 2026, Dry Lab | Layer 0 of 17-Layer Biosensor Modelling Pipeline | `wound_biology.py`

Purpose

Layer 0 predicts the **free protease concentration in wound fluid at the time of swab sampling** and how it evolves over the 2-hour post-sampling window. Its outputs, $[\text{MMP-9}]_{\text{free}}(t)$ and $[\text{HNE}]_{\text{free}}(t)$, are in-situ wound fluid concentrations. Layer 1 (sample collection) applies the swab dilution factor D_f to convert from in-situ concentrations to what enters the device.

The distinction between healing and chronic wound states is encoded through two distinct initial-condition parameter sets, both grounded in published ELISA data. The model is a system of ODEs for nine coupled species: pro-MMP-9, free MMP-9, free TIMP-1, MMP-9:TIMP-1 complex, free HNE, free A1AT, HNE:A1AT complex, free SLPI, and HNE:SLPI complex.

Parameter citation policy. Every parameter used by this model must be grounded in a paper in `electrochemical_device/papers/layer_00/` (status: ✓), approximate from a figure in such a paper (status: ∼), or explicitly flagged as awaiting measurement (status: ○). No values from secondary literature alone.

I. Wound Fluid Composition

Biological rationale

Chronic wounds are characterised by a prolonged inflammatory phase driven by persistent neutrophil and macrophage infiltration[1]. This sustains elevated protease secretion long after acute wounds have resolved. The two proteases targeted by this device are:

- **MMP-9 (gelatinase B)**: secreted by neutrophils and macrophages as an inactive zymogen (pro-MMP-9). Activation requires proteolytic removal of the prodomain. In chronic wounds, both forms are elevated; the ratio of active to pro form predicts healing outcome.
- **HNE (human neutrophil elastase)**: a serine protease released already active from azurophil granules. Regulated primarily by α_1-antitrypsin (A1AT) and secondarily by SLPI.

Literature evidence

Trengove et al. (1999)[1] measured protease activity in wound fluid from chronic venous leg ulcers (VLU) versus acute wounds (all values from paper text, p.446–448):

- Chronic VLU MMP activity: median 22.8 μg/mL, versus 0.76 μg/mL in acute wounds ($\approx 30\times$ elevation, $p < 0.001$; 92-kDa MMP-9 predominant on zymography).
- TIMP-1 (ELISA): chronic VLU median 0.8 μg/mL = 800 ng/mL ($n = 5$; text p.448).
- HNE (activity assay): non-healing VLU median 1.4 μg/mL $\approx$ 47 nM (range < 1–11.9 μg/mL; elevated in only 6 of 13 chronic samples; text p.448).
- MMP activity decreased significantly as wounds healed ($p < 0.01$; text p.449).

Ladwig et al. (2002)[2] used quantitative gelatin zymography and ELISA in wound fluid from 56 patients with chronic pressure ulcers. Key values (all from paper text p.31):

- Pro-MMP-9: mean 8,100 ± 3,000 ng/mL. Active MMP-9: mean 4,800 ± 1,300 ng/mL.
- Total MMP-9 = 8,100 + 4,800 = 12,900 ng/mL; pro-form fraction = 8,100/12,900 = 63%.
- Active MMP-9 significantly higher in poor healers at Day 0 (text p.33; approximate values from Figure 8).
- TIMP-1 significantly higher in good healers at Day 0 (text p.33; approximate values from Figure 10).
- The MMP-9:TIMP-1 ratio was a predictor of healing trajectory (text p.34, Figure 11).

Rao et al. (1995)[7] examined A1AT integrity in wound fluid from chronic and healing wounds (J Invest Dermatol 105:572–578, text p.572–575). In healing and acute wound fluid, A1AT was intact and functional in all samples. In chronic wound fluid, A1AT was degraded and non-functional in 9 of 10 samples; the authors attribute this to sustained serine and metalloprotease activity in the chronic wound environment. This finding is critical for the HNE channel: functional A1AT cannot be assumed present in chronic wounds.

Nwomeh et al. (1999)[8] measured matrix metalloproteinase activities and inhibitor levels in venous leg ulcers (J Surg Res 81:189–195). TIMP-1 levels in chronic ulcer fluid were "very low or not detected" (text p.193), consistent

with the low TIMP-1 values in Trengove 1999[1] and confirming TIMP-1 suppression as a hallmark of the chronic wound environment.

Outstanding measurements:

Healing wound concentrations (not confirmed): Ladwig 2002 is a pressure ulcer study; the healing scenario MMP-9 (≈ 5,400 ng/mL) is read approximately from Figure 8 (not stated in text). The healing TIMP-1 (= 1,104 ng/mL) has no paper-text source – it is a working estimate. The healing wound prediction (Section V) is therefore illustrative. It depends on the sign of the MMP-9/TIMP-1 excess, not the exact values; a confirmed joint VLU ELISA (MMP-9 + TIMP-1 in the same cohort, chronic and healing time points) is needed before the discrimination ratio is finalised.

Initial conditions

Status: ✓ confirmed from paper text, ∼ approximate from figure, ∘ needs measurement.

Parameter	Chronic wound	Healing wound	St.	Source
$[\text{MMP-9}]_{\text{total}}$	12,900 ng/mL (140 nM)	5,400 ng/mL (59 nM)	✓/∼	[2] text/Fig. 8
[pro-MMP-9] (63%)	8,100 ng/mL (88 nM)	3,400 ng/mL (37 nM)	✓/∼	[2] text
[active MMP-9] (37%)	4,800 ng/mL (52 nM)	2,000 ng/mL (22 nM)	✓/∼	[2] text/Fig. 8
[TIMP-1]	800 ng/mL (28 nM)	1,104 ng/mL (39 nM)	✓/∘	[1] text/estimate
[HNE] (both states)	47 nM (1.4 μg/mL)		✓	[1] text
[A1AT] (functional)	≈ 0 (degraded)	1.92 μM (estimate)	✓/∘	[7] text/est.

Physical basis for wound-state discrimination. In chronic wounds, active MMP-9 (52 nM) exceeds TIMP-1 (28 nM) by 24 nM. Since K_i(TIMP-1/MMP-9) = 8.5 nM (Olson 1997[4], confirmed), which is small compared to both concentrations, TIMP-1 is nearly fully bound and free MMP-9 persists at ≈ 32 nM (ODE result, t = 30 min). In healing wounds (estimated), TIMP-1 (39 nM) exceeds active MMP-9 (22 nM) by 17 nM, suppressing free MMP-9 to ≈ 6 nM.

A1AT integrity provides a second discriminating axis for HNE (Rao 1995[7]): in healing wounds A1AT is intact, quenching HNE to 0.00 pM within milliseconds; in chronic wounds A1AT is degraded in 9/10 samples, leaving HNE partially regulated only by SLPI (≈30 nM at $t = 30$ min). However, elevated HNE was detected in only 6 of 13 chronic VLU samples (Trengove 1999[1]), limiting its reliability as a primary discriminator.

II. Activation Kinetics (MMP-9 Pathway)

Pro-MMP-9 activation by MMP-3 and plasmin is modelled as pseudo-first-order in pro-MMP-9:

$$\frac{d[\text{pro-MMP-9}]}{dt} = -\left(\frac{k_{\text{cat}}}{K_m}\right)_{\text{MMP3}} [\text{MMP-3}]\,[\text{pro-MMP-9}] - \left(\frac{k_{\text{cat}}}{K_m}\right)_{\text{plasmin}} [\text{plasmin}]\,[\text{pro-MMP-9}] \tag{1}$$

Outstanding measurements:

$(k_{\text{cat}}/K_m)_{\text{MMP3}}$, $(k_{\text{cat}}/K_m)_{\text{plasmin}}$, [MMP-3], and [plasmin] in wound fluid are all unconfirmed placeholders. No paper in the papers folder provides wound-fluid MMP-3 or plasmin concentrations or the relevant kinetic constants. With current placeholder values, $< 1\%$ of pro-MMP-9 is activated in 120 min, so these values have no effect on the chronic wound free MMP-9 prediction. They would matter only if pro-MMP-9 activation within the 2-hour window were a significant fraction of the total.

III. Inhibition Kinetics

TIMP-1 / MMP-9 (reversible tight inhibition)

TIMP-1 binds to the MMP-9 active site in a non-covalent, reversible manner with biphasic kinetics (two binding sites; high-affinity site dominates at wound fluid concentrations):

$$\text{MMP-9} + \text{TIMP-1} \underset{k_{\text{off}}}{\overset{k_{\text{on}}}{\rightleftharpoons}} \text{MMP-9:TIMP-1} \qquad K_i = \frac{k_{\text{off}}}{k_{\text{on}}} \tag{2}$$

Confirmed kinetic constants (Olson et al. 1997[4], Table IV, enzyme inhibition assay, MMP-9 82 kDa active form):

$k_{\text{on}} = 2.48 \pm 0.47 \times 10^5\ \text{M}^{-1}\text{s}^{-1}$
$k_{\text{off}} = 2.1 \pm 1.1 \times 10^{-3}\ \text{s}^{-1}$
$K_i = k_{\text{off}}/k_{\text{on}} = 8.5 \pm 2.9\ \text{nM}$ (stated directly in Table IV; check: $2.1 \times 10^{-3}/2.48 \times 10^5 = 8.47\ \text{nM}$)

Enzyme inhibition assay values are used in preference to SPR values (Olson 1997 Tables I and II) because they reflect solution-phase kinetics at conditions closer to wound fluid. As noted by Olson 1997 (p.29981), enzyme inhibition k_{on} values are consistently 2.5–7× higher than SPR values, making them more appropriate for modelling diffusion-limited association in wound fluid.

Sensitivity of discrimination ratio to K_i. The chronic wound prediction (≈ 32 nM free MMP-9 at 30 min) is robust: changing K_i from 8.5 nM to 1 nM shifts the result from 32 nM to 27 nM (a 15% change). The healing prediction is more sensitive: $K_i = 8.5$ nM gives 6 nM free MMP-9, whereas $K_i = 1$ nM gives 1.3 nM. This is because the healing wound MMP-9/TIMP-1 excess is only 17 nM, comparable to K_i.

A1AT / HNE (near-irreversible inhibition)
A1AT forms a covalent acyl-enzyme intermediate with HNE that is effectively irreversible:

$$\text{HNE} + \text{A1AT} \xrightarrow{k_{\text{on}}} \text{HNE:A1AT} \quad (\text{irreversible};\ k_{\text{off}} \approx 0) \tag{3}$$

Confirmed kinetic constant (Beatty, Bieth & Travis 1980[5], Table 1):

$$k_{\text{on}}(\text{HNE/A1AT}) = 6.5 \pm 4.0 \times 10^7\ \text{M}^{-1}\text{s}^{-1} \quad (\text{at } 25\,^\circ\text{C, pH } 8.0) \tag{4}$$

Half-life of free HNE in healing wounds (with functional $[\text{A1AT}]_0 = 1.92\ \mu\text{M}$, estimated):

$$t_{1/2}^{\text{HNE}} = \frac{\ln 2}{k_{\text{on}} \cdot [\text{A1AT}]_0} = \frac{0.693}{6.5 \times 10^7 \times 1.92 \times 10^{-6}} = 0.0055\ \text{s} \quad (5.5\ \text{ms}) \tag{5}$$

Because $[\text{A1AT}]_0$ is an estimate, the conclusion is verified at 10× lower $[\text{A1AT}]_0$: $t_{1/2} = 55$ ms, still three orders of magnitude below the 5-minute measurement window. This analysis applies to healing wounds where A1AT is intact (Rao et al. 1995[7]). In chronic wounds, A1AT is degraded in 9 of 10 samples; see the finding below.

Critical finding:

A1AT is wound-state dependent (Rao et al. 1995[7]). In **healing** wounds A1AT is intact; HNE is quenched to 0.00 pM within 5.5 ms (confirmed $k_{on} = 6.5 \times 10^7$ $M^{-1}s^{-1}$, Beatty 1980[5]; insensitive to [A1AT] estimate: $t_{1/2} = 55$ ms even at 10× lower A1AT). In **chronic** wounds, A1AT is degraded and non-functional in 9/10 samples (Rao 1995[7]); HNE is only partially inhibited by SLPI, persisting at ≈30 nM at $t = 30$ min.
HNE is an unreliable primary sensor target. Although HNE discriminates between wound states (chronic ≈30 nM vs healing 0 pM), elevated HNE was detected in only 6 of 13 chronic VLU samples (Trengove 1999[1]): the remaining 50% of chronic cases have HNE at or below detection. MMP-9 is the preferred primary target. The second electrode should serve as an internal reference or control, not a primary HNE sensing channel.

SLPI / HNE (secondary inhibitor)

SLPI is a serine protease inhibitor expressed in wound healing tissue[6]. Placeholder kinetic constants ($k_{on} = 10^5$ $M^{-1}s^{-1}$, $k_{off} = 10^{-4}$ s^{-1}, concentration ≈200 ng/mL ≈ 17 nM) are used. Ashcroft et al. (2000)[6] confirmed the importance of SLPI to wound healing outcomes in knockout mice but did not state a quantitative SLPI concentration in human wound fluid.

In healing wounds, HNE is completely neutralised by intact A1AT (Rao 1995[7]), making SLPI irrelevant. In chronic wounds, A1AT is degraded in 9/10 samples (Rao 1995[7]), and SLPI provides the primary remaining HNE inhibition; the ODE predicts ≈30 nM free HNE at $t = 30$ min after partial SLPI inhibition. The chronic wound HNE prediction is sensitive to placeholder SLPI kinetics and concentration, which remain unconfirmed from papers in this folder.

IV. Full ODE System

Variable key (all concentrations in mol/L):

P	pro-MMP-9	C_1	MMP-9:TIMP-1 complex
M	free (active) MMP-9	H	free HNE
T	free TIMP-1	A	free A1AT
C_2	HNE:A1AT complex	S	free SLPI
C_3	HNE:SLPI complex		

$$\dot{P} = -k^* P \tag{6}$$

$$\dot{M} = +k^* P - k^T_{on} M T + k^T_{off} C_1 \tag{7}$$

$$\dot{T} = -k^T_{on} M T + k^T_{off} C_1 \tag{8}$$

$$\dot{C}_1 = +k^T_{on} M T - k^T_{off} C_1 \tag{9}$$

$$\dot{H} = -k^A_{on} H A - k^S_{on} H S + k^S_{off} C_3 \tag{10}$$

$$\dot{A} = -k^A_{on} H A \tag{11}$$

$$\dot{C}_2 = +k^A_{on} H A \tag{12}$$

$$\dot{S} = -k^S_{on} H S + k^S_{off} C_3 \tag{13}$$

$$\dot{C}_3 = +k^S_{on} H S - k^S_{off} C_3 \tag{14}$$

where $k^* = (k_{cat}/K_m)_{MMP3}[\text{MMP-3}] + (k_{cat}/K_m)_{plasmin}[\text{plasmin}]$; $k^T_{on,off}$ for TIMP-1/MMP-9 (Olson 1997[4]); k^A_{on} for A1AT/HNE (Beatty 1980[5]).

Solved with the Radau stiff solver (`scipy.integrate.solve_ivp`, `rtol=1e-10, atol=1e-15`). Stiff integration is required because the A1AT/HNE timescale (∼5.5 ms) is 10^7-fold faster than the pro-MMP-9 activation timescale (∼30–120 min). The TIMP-1/MMP-9 equilibration is intermediate ($\tau \approx 2$ min).

V. Model Output and Wound-State Discrimination

Discrimination between wound states at $t = 30$ min

Parameter	Chronic wound	Healing wound	C/H	Source
$[\text{MMP-9}]_{\text{total}}$ (in-situ)	12,900 ng/mL (140 nM)	5,400 ng/mL (59 nM)	2.4×	[2] text/Fig.
$[\text{MMP-9}]_{\text{active}}$ (in-situ)	4,800 ng/mL (52 nM)	2,000 ng/mL (22 nM)	2.4×	[2] text/Fig.
[TIMP-1] (in-situ)	800 ng/mL (28 nM)	1,104 ng/mL (39 nM)	0.72×	[1] text/est.
MMP-9 vs TIMP-1 excess	+24 nM (MMP-9)	−17 nM (TIMP-1)	–	derived
ODE output at $t = 30$ min (in-situ, before Layer 1 dilution D_f):				
$[\text{MMP-9}]_{\text{free}}$	**32** nM	**6** nM	**5**×	ODE
$[\text{TIMP-1}]_{\text{free}}$	≈6 nM	≈22 nM	–	ODE
$[\text{HNE}]_{\text{free}}$	≈30 nM†	0.00 pM	–	ODE[7]

† A1AT degraded in chronic wounds (Rao 1995[7]); HNE elevated in only 50% of chronic VLU (Trengove 1999[1]). Chronic HNE ≈30 nM is sensitive to placeholder SLPI kinetics.

Confirmed chronic wound prediction. With Ladwig 2002 text-confirmed concentrations (active MMP-9 = 52 nM, TIMP-1 = 28 nM)[2] and Olson 1997 confirmed kinetics ($K_i = 8.5$ nM)[4], the ODE gives free MMP-9 ≈ 32 nM at $t = 30$ min in chronic wounds. This is driven entirely by the 24 nM MMP-9 excess over TIMP-1; no production rate assumptions are needed.

Illustrative healing comparison. The healing result (≈ 6 nM) depends on unconfirmed estimates: healing MMP-9 from Ladwig Figure 8 (approximate) and healing TIMP-1 with no paper source. The discrimination ratio (∼5×) should be treated as a working estimate until a joint VLU ELISA is performed. The mechanism is sound: when TIMP-1 excess exceeds active MMP-9, free MMP-9 is suppressed; the exact ratio follows from the confirmed K_i.

HNE secondary channel. Confirmed A1AT wound-state dependence (Rao 1995[7]) gives strong HNE discrimination: chronic ≈30 nM vs healing 0.00 pM. However, elevated HNE was observed in only 6/13 (50%) of chronic VLU samples (Trengove 1999[1]); the other 50% show HNE at or below detection. HNE is therefore not a reliable standalone discriminator and the 30 nM chronic estimate is sensitive to unconfirmed SLPI kinetics. MMP-9 (∼5× discrimination, grounded in confirmed kinetics) is the preferred primary target.

Layer 1 dilution and sensor range

For chronic wound free MMP-9 (32 nM) to land in the sensor range (17–200 ng/mL = 0.185–2.17 nM):

$$D_f = \frac{[\text{MMP-9}]_{\text{free, in-situ}}}{[\text{MMP-9}]_{\text{target}}} = \frac{32 \text{ nM}}{0.185\text{–}2.17 \text{ nM}} \approx 15\text{–}173 \tag{15}$$

A dilution factor $D_f \approx 10$–100 is within the range of practical swab designs. Example with $D_f = 50$: chronic wound device entry = 32/50 = 0.64 nM = 59 ng/mL (within sensor range); healing wound = 6/50 = 0.12 nM = 11 ng/mL (near the lower end). The healing signal approaching the LOD under large D_f is a risk to address in Layer 1.

VI. Parameter Summary

Status: ✓ confirmed from paper text, ∼ from figure (not text), ∘ needs measurement/citation. **Sensitivity**: whether changing the value by 10× affects the device-relevant predictions.

Parameter	Value	Source	St.	Affects	Sensitv.
$[\text{MMP-9}]_{\text{pro}}$ (chr.)	8,100 ng/mL	Ladwig 2002[2] text p.31	✓	I.C.	Low
$[\text{MMP-9}]_{\text{act}}$ (chr.)	4,800 ng/mL	Ladwig 2002[2] text p.31	✓	I.C.	High
Pro-form fraction	63%	Ladwig 2002[2] (8100/12900)	✓	I.C.	Low
[TIMP-1] (chronic)	800 ng/mL	Trengove 1999[1] text p.448	✓	MMP-9 eq.	High
[HNE]	47 nM	Trengove 1999[1] text p.448	✓	HNE (n/v)	n/a
$[\text{MMP-9}]_{\text{act}}$ (heal.)	2,000 ng/mL	Ladwig 2002[2] Fig. 8 (approx.)	∼	I.C. (heal.)	High
[TIMP-1] (healing)	1,104 ng/mL	No paper in folder – estimate	○	MMP-9 eq.	High
k_{on} (TIMP-1/MMP-9)	2.48×10^5 $\text{M}^{-1}\text{s}^{-1}$	Olson 1997[4] Table IV	✓	MMP-9 eq.	Low
k_{off} (TIMP-1/MMP-9)	2.1×10^{-3} s^{-1}	Olson 1997[4] Table IV	✓	K_i	Moderate
K_i (TIMP-1/MMP-9)	8.5 nM	Olson 1997[4] Table IV	✓	MMP-9 eq.	Moderate
k_{on} (A1AT/HNE)	6.5×10^7 $\text{M}^{-1}\text{s}^{-1}$	Beatty 1980[5] Table 1	✓	HNE (heal.)	Low
[A1AT] (healing)	1.92 μM (est.)	Serum ÷10; intact (Rao 1995[7])	✓/○	HNE 0 pM	Low*
[A1AT] (chronic)	≈ 0 (degraded)	Rao 1995[7] text: 9/10 samples	✓	HNE ≈30 nM	High
[MMP-3]	10 ng/mL	Placeholder	○	Act. rate	None
[plasmin]	1 nM	Placeholder	○	Act. rate	None
k_{cat}/K_m (MMP-3)	5×10^3 $\text{M}^{-1}\text{s}^{-1}$	Placeholder	○	Act. rate	None
[SLPI]	200 ng/mL	Ashcroft 2000[6]: no conc. stated	○	HNE (chr.)	Moderate

**In healing wounds, A1AT intact (Rao 1995[7]): HNE $t_{1/2} = 55$ ms even at 10× lower [A1AT]; healing HNE = 0.00 pM is insensitive to this estimate. In chronic wounds, A1AT ≈ 0 (confirmed degraded); chronic HNE prediction depends on placeholder SLPI kinetics (Moderate sensitivity).*

References

All papers in `electrochemical_device/papers/layer_00/` unless otherwise noted.

[1] Trengove NJ, Stacey MC, Macauley S et al. (1999) Analysis of the acute and chronic wound environments: the role of proteases and their inhibitors. *Wound Repair Regen* 7(6):442–52. **trengove1999.pdf** – values from text confirmed.

[2] Ladwig GP, Robson MC, Liu R et al. (2002) Ratios of activated matrix metalloproteinase-9 to tissue inhibitor of metalloproteinase-1 in wound fluids are inversely correlated with healing of pressure ulcers. *Wound Repair Regen* 10(1):26–37. **ladwig2002.pdf** – chronic concentrations from text (p.31) confirmed. Healing concentrations from Figure 8 only (approximate).

[3] Nagase H, Woessner JF Jr (1999) Matrix metalloproteinases. *J Biol Chem* 274(31):21491–4. **nagase_woessner1999.pdf** – MMP overview. Contains no kinetic constants for TIMP-1/MMP-9; cited for biological context only.

[4] Olson MW, Gervasi DC, Mobashery S, Fridman R (1997) Kinetic analysis of the binding of human matrix metalloproteinase-2 and -9 to tissue inhibitor of metalloproteinase (TIMP)-1 and TIMP-2. *J Biol Chem* 272(47):29975–83. **olson1997.pdf** – k_{on}, k_{off}, K_i for MMP-9 (82 kDa)/TIMP-1 confirmed from Table IV (enzyme inhibition assay).

[5] Beatty K, Bieth J, Travis J (1980) Kinetics of association of serine proteinases with native and oxidized α-1-proteinase inhibitor and α-1-antichymotrypsin. *J Biol Chem* 255(9):3931–34. **beatty1980.pdf** – $k_{\text{on}}(\text{HNE}/\alpha_1\text{-PI}) = 6.5 \times 10^7$ $\text{M}^{-1}\text{s}^{-1}$ confirmed from Table 1.

[6] Ashcroft GS, Lei K, Jin W et al. (2000) Secretory leukocyte protease inhibitor mediates non-redundant functions necessary for normal wound healing. *Nat Med* 6(10):1147–53. **ashcroft2000.pdf** – confirms SLPI role in wound healing (knockout mouse model). Does not provide SLPI concentration in human wound fluid; SLPI concentration remains unconfirmed in this model.

[7] Rao CN, Ladin DA, Liu YY et al. (1995) α_1-Antitrypsin is degraded and non-functional in chronic wounds but intact and functional in acute wounds: the inhibitor protects fibronectin from degradation by chronic

wound fluid enzymes. *J Invest Dermatol* 105(4):572–578. **rao1995.pdf** – A1AT intact in all healing/acute samples; degraded and non-functional in 9/10 chronic wound samples. Confirmed from text (p.572–575). No quantitative A1AT concentration in ng/mL is provided; healing-state concentration remains an estimate (serum ÷10).

[8] Nwomeh BC, Liang HX, Cohen IK, Yager DR (1999) MMP-8 is the predominant collagenase in healing wounds and nonhealing ulcers. *J Surg Res* 81(2):189–195. **nwomeh1999.pdf** – TIMP-1 levels "very low or not detected" in chronic ulcer fluid (text p.193), confirming TIMP-1 suppression in chronic wound environments. Quantitative TIMP-1 values are in ng per 100 μg protein (not ng/mL); cannot be converted to absolute concentration without protein content data.